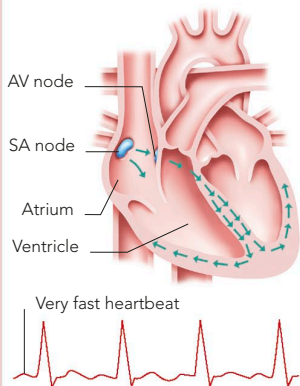


## ARRHYTHMIA

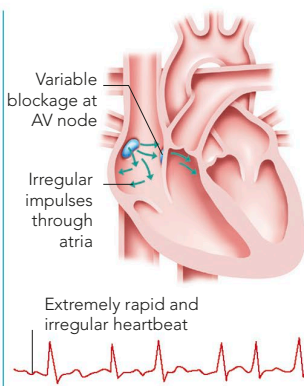
An arrhythmia is a heart rate that is unusually slow or fast, or erratic. A normal heartbeat is initiated by specialized cells in the natural “pacemaker”, the sinoatrial (SA) node, at the top of the right atrium. They send electrical impulses out through the atrial muscle tissue, stimulating it to contract.

These signals are relayed by the atrioventricular (AV) node along nerve-like fibres through the septum (central dividing wall) and into the thick muscle tissue of the ventricle walls. A fault in the system can lead to various arrhythmias, including those described here.



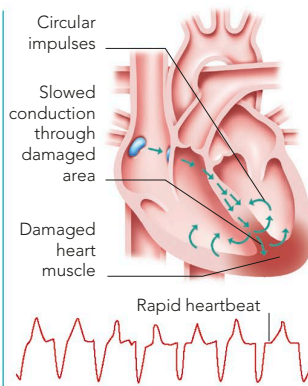
### SINUS TACHYCARDIA

In sinus tachycardia there is a regular but rapid heart rate, usually more than 100 beats per minute.



### ATRIAL FIBRILLATION

Fibrillations are rapid, disordered, weak contractions with a rate as high as 500 per minute.



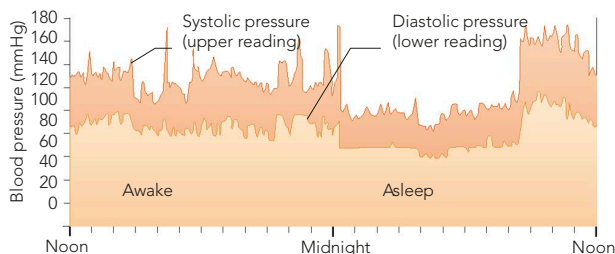
### VENTRICULAR TACHYCARDIA

Very fast ventricular contractions may be caused by heart muscle damage, for example due to a heart attack.

## HYPERTENSION

Blood is under pressure as the heart pumps it around the circulation. In hypertension, this pressure is persistently above normal limits. There are no symptoms of hypertension at first, but despite this, over time it increases the risk of many serious disorders, such as stroke, heart disease, and kidney failure. Contributing factors

to hypertension include certain genetic influences and diet and lifestyle factors, such as being overweight, drinking excessive amounts of alcohol, smoking, and having a high-salt diet. Hypertension is most common in middle-aged and elderly people. A stressful lifestyle may aggravate the condition.



### BLOOD PRESSURE GRAPH

Normal blood pressure varies according to activity levels. This graph shows that during sleep, both the systolic and diastolic pressures (see pp.154–155) are much lower.